EXP2: Latest Results vs. Manuscript

Latest run: 2026-11-06 01:47:25 (UTC+08:00; saved 01:48:02).

256×256; simulated 3 ms, evaluated 3 ms; 16 temporal bins; seed 42.

RMSE (px/ms)

	JHTDB-isotropic		DNS-turbulence		SQG	
Method	Latest	Manuscript	Latest	Manuscript	Latest	Manuscript
ILK-VG	0.6154	0.6116	0.4414	1.1221	0.6852	0.6809
OF-VG	0.4948	0.4869	0.3453	0.3441	0.7624	0.5798
INR-EST	0.6110	0.6565	0.3823	0.4234	0.5506	0.5764
INR-VG	0.5628	0.5529	0.3767	0.3832	0.6113	0.6035
ConstGS	0.2842	0.3324	0.1774	0.1744	0.3892	0.4468
AffineGS	0.1617	0.1780	0.1272	0.1299	0.2188	0.2294

AEE (px/ms)

	JHTDB-isotropic		DNS-turbulence		SQG	
Method	Latest	Manuscript	Latest	Manuscript	Latest	Manuscript
ILK-VG	0.4599	0.4538	0.3640	0.3658	0.5328	0.5268
OF-VG	0.3788	0.3724	0.2866	0.2844	0.5057	0.4581
INR-EST	0.4633	0.5006	0.3284	0.3635	0.4472	0.4626
INR-VG	0.4221	0.4137	0.3187	0.3235	0.4936	0.4862
ConstGS	0.2204	0.2554	0.1354	0.1366	0.2959	0.3386
AffineGS	0.1287	0.1423	0.0989	0.1038	0.1677	0.1834

EXP2: Flow-gradient Metrics

Lower is better. Units: ms^{-1}.

RMSE-div ($\mathbf{ms}^{-1}$)

	JHTDB-isotropic		DNS-turbulence		SQG	
Method	Latest	Manuscript	Latest	Manuscript	Latest	Manuscript
ILK-VG	0.0579	0.0529	0.0538	0.4264	0.0675	0.0605
OF-VG	0.0526	0.0450	0.0462	0.0398	0.0758	0.0558
INR-EST	0.0216	0.0230	0.0159	0.0195	0.0316	0.0341
INR-VG	0.0236	0.0238	0.0184	0.0186	0.0319	0.0335
ConstGS	0.0165	0.0141	0.0092	0.0087	0.0281	0.0329
AffineGS	0.0083	0.0098	0.0174	0.0186	0.0193	0.0188

RMSE-ω ($\mathbf{ms}^{-1}$)

	JHTDB-isotropic		DNS-turbulence		SQG	
Method	Latest	Manuscript	Latest	Manuscript	Latest	Manuscript
ILK-VG	0.1464	0.1437	0.0955	0.7369	0.1767	0.1742
OF-VG	0.1352	0.1240	0.0829	0.0746	0.1683	0.1492
INR-EST	0.1484	0.1535	0.0852	0.0888	0.1548	0.1565
INR-VG	0.1411	0.1400	0.0846	0.0850	0.1655	0.1646
ConstGS	0.0919	0.1004	0.0516	0.0515	0.1157	0.1258
AffineGS	0.0663	0.0700	0.0461	0.0485	0.0802	0.0838

Latest: `output/EXP2/on_single_flow/table1-velocity-metrics.csv` and `table2-gradient-metrics.csv`. Manuscript: Tables 1 and 2 of `manuscript_revised.pdf`.